Area of Triangles

Lesson 5-3

Name: ________ Date: ________ Class: ________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

If the bottom of a triangle is 10 cm, then b = 10 cm in the formula $A = \frac{1}{2} \times b \times h$

Base

A side of the triangle you use to find the area.

A dashed vertical line from the top point straight down to the base at a 90° angle — like dropping a plumb line

Height

The straight-up distance from the base to the top corner.

A triangle with b = 8 and h = 6 has area = $\frac{1}{2} \times 8 \times 6 = 24$ sq units — exactly half the rectangle around it

Area

How much space is inside a flat shape.

The corner of a book or a door frame — the edges meet at exactly 90°, shown by a small square symbol

Perpendicular

Two lines that meet to make a square corner (90 degrees).

A house shape = a rectangle (the walls) + a triangle (the roof); total area = rectangle area + triangle area

Composite figure

A shape made by putting two or more simple shapes together.

$A = \frac{1}{2} \times b \times h$ means Area equals one-half times base times height; for b = 12 and h = 8, A = 48

Formula

A math rule written with symbols.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the area of a triangle using the formula A = ½ × base × height.

1 Notice

Your architecture firm is designing a park with triangular garden beds.

2 Key idea

I can find the area of a triangle using the formula A = ½ × base × height.

3 Words I'll use

Base **Height** **Area** **Perpendicular** **Composite figure** **Formula**

🤔 Think About It

- What shape are the garden beds?
- What measurements do we need to find the area?
- How is the area of a triangle related to the area of a rectangle?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The triangular garden bed has a base of 12 feet and a height of 8 feet. Which measurement is the height, and how is it different from the slanted side?

Level 1 support

Start here: The height is the perpendicular distance from the base to the top corner.

💡 **Need a hint?**

Sentence starters (optional)

The height is the ____ distance from the base to the top corner.

La altura es la distancia ____ desde la base hasta la esquina de arriba.

It is different from the slanted side because ____.

Es diferente del lado inclinado porque ____.

Word bank: base | height | perpendicular | area | formula

Listen for: *Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.*

Level 2

Does it matter which side of the garden triangle you call the base? Justify your thinking using the formula A = 1/2 x b x h.

- It ____ matter which side is the base because ____.
- No matter which base I choose, the area is the same since ____.

EXPLORE

On the grid your triangle (base 12, height 8) fits inside a 12 by 8 rectangle. Why do we divide by 2 to find the triangle's area?

Level 1 support

Start here: A triangle is exactly half of a rectangle with the same base and height.

💡 **Need a hint?**

Sentence starters (optional)

A triangle is half of a ___ with the same base and height.

Un triángulo es la mitad de un ___ con la misma base y altura.

So the area is 1/2 times 12 times 8, which equals ___.

Entonces el área es 1/2 por 12 por 8, que es ___.

The rectangle's area is ___, so the triangle is ___.

El área del rectángulo es ___, entonces el triángulo es ___.

Word bank: base height area perpendicular formula

Listen for: *A strong answer states the triangle is half of the 12 x 8 = 96 rectangle, so the triangle area is 48 sq ft, and connects the 1/2 to that halving.*

Level 2

Two different triangles both have area 48 sq ft. Must they have the same base and height? Justify or give a counterexample.

- They ___ have the same base and height because ___.
- A counterexample is a triangle with base ___ and height ___, which also gives ___.

CONNECT

The homeowner paints a triangular A-frame wall with base 20 ft and height 14 ft; one can covers 50 sq ft. Talk through how many cans are needed.

Level 1 support

Start here: Find the triangle area first, then divide by what one can covers.

💡 **Need a hint?**

Sentence starters (optional)

The wall's area is 1/2 times 20 times 14, which equals ___ square feet.

El área de la pared es 1/2 por 20 por 14, que es ___ pies cuadrados.

She needs ___ cans because ___ divided by 50 equals ___.

Necesita ___ latas porque ___ entre 50 es ___.

Word bank: triangle | base | height | area | formula

Listen for: *Students compute area = 140 sq ft, then 140 / 50 = 2.8, and reason she must round up to 3 cans to cover the whole wall.*

Level 2

Why can't the homeowner buy exactly 2.8 cans? Explain what rounding up means in this real situation and how it changes the leftover paint.

- She must round up to ___ cans because ___.
- The leftover paint covers about ___ sq ft, since ___.

Guided Examples

Example 1

What is the area of a triangle with base 10 cm and height 6 cm?

Solution: $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 10 \times 6 = 30$ square centimeters.

Answer: A. 30 sq cm

Example 2

A triangle has an area of 24 sq ft and a base of 8 ft. What is the height?

Solution: $A = \frac{1}{2} \times b \times h \rightarrow 24 = \frac{1}{2} \times 8 \times h \rightarrow 24 = 4h \rightarrow h = 6$ ft.

Answer: A. 6 ft

Example 3

Why do we divide by 2 when finding the area of a triangle?

Solution: A triangle with base b and height h fits inside a rectangle with the same base and height. The triangle covers exactly half the rectangle, so $A = \frac{1}{2} \times b \times h$.

Answer: A. A triangle is exactly half of a rectangle with the same base and height

Write About the Math
The Writing Revolution

I can explain my steps using the words base, height, area, and perpendicular.

1. Kernel Sentence
subject + verb

Model: Area is how much space is inside a flat shape.
Área es cuánto espacio hay dentro de una figura plana.

Write a kernel sentence about area. Use a subject and a verb.
Escribe una oración base sobre área. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion
because · but · so

Kernel: Area matters in math
Área importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Area matters in math because ____.
Área importa en matemáticas porque ____.

___

but
pero

Area matters in math, but ____.
Área importa en matemáticas, pero ____.

___

so
entonces

Area matters in math, so ____.
Área importa en matemáticas, entonces ____.

___

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about area.
Di un hecho verdadero sobre area.

Area ____.

Question
Pregunta

Ask a question about area.
Haz una pregunta sobre area.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about area.
Muestra entusiasmo sobre area.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with area.
Dile a un compañero qué hacer con area.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

The height is the ____ distance from the base to the top corner.
La altura es la distancia ____ desde la base hasta la esquina de arriba.

It is different from the slanted side because ____.
Es diferente del lado inclinado porque ____.

A triangle is half of a ____ with the same base and height.
Un triángulo es la mitad de un ____ con la misma base y altura.

Try It

Solve on your own. Check the answer key when you are done.

1. Why do we divide by 2 when finding the area of a triangle?

A. A triangle is exactly half of a rectangle with the same base and height

B. Triangles have 2 equal sides

C. The base is always twice the height

D. We always divide area formulas by 2

Show your work:

2. What is the area of a triangle with base 14 ft and height 6 ft?

A. 42 sq ft

B. 84 sq ft

C. 20 sq ft

D. 42 ft

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A rectangle is 12 ft by 8 ft. A diagonal line cuts it into two triangles. What is the area of each triangle? Explain how the triangle area formula relates to the rectangle area formula.

Sentence starter: The rectangle's area is ____ because ____. Each triangle's area is ____ because a diagonal splits a rectangle into ____ equal triangles, so A = ½ × ____ × ____ = ____.

Show your work:

Reflect — Exit Ticket

A triangle has a base of 11 inches and a height of 8 inches. What is its area?

A. 44 sq in
B. 88 sq in
C. 19 sq in
D. 44 in

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. A triangle is exactly half of a rectangle with the same base and height *— A triangle with base b and height h fits inside a rectangle with the same base and height. The triangle covers exactly half the rectangle, so $A = ½ \times b \times h$.*
2. **Try It 2:** A. 42 sq ft *— $A = ½ \times 14 \times 6 = ½ \times 84 = 42$ sq ft.*
3. **Exit Ticket:** A. 44 sq in *— $A = 1/2 \times 11 \times 8 = 1/2 \times 88 = 44$ square inches.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Area is how much space is inside a flat shape.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).